

A. Real magnitude baseline

4-channel magnitude input



[R-Block + MaxPool] $\times 3$



R-Block



Global AvgPool



Dropout



Linear logit

B. Complex-only family

4-channel complex input



[C-Block + C-Pool] $\times 3$



Optional Holo-
graphic Attention



C-Block



magnitude readout $|\cdot|$



Global AvgPool



Dropout



Linear logit

C. Dual-stream family

4-channel complex input



magnitude $|\mathbf{x}|$

complex stream $\mathbf{x}$

[R-Block +
MaxPool] $\times 3$

Modulus
gate
(if selected)

[C-Block +
C-Pool] $\times 3$



R-Block

Real features

Global AvgPool



Concatenate



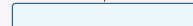
Dropout



Linear logit



Optional
holographic
attention



C-Block

Complex features

$|\cdot|$



Global AvgPool



Complex-model channel widths are adjusted per configuration to match the fixed real-baseline scalar parameter budget within 1%.